



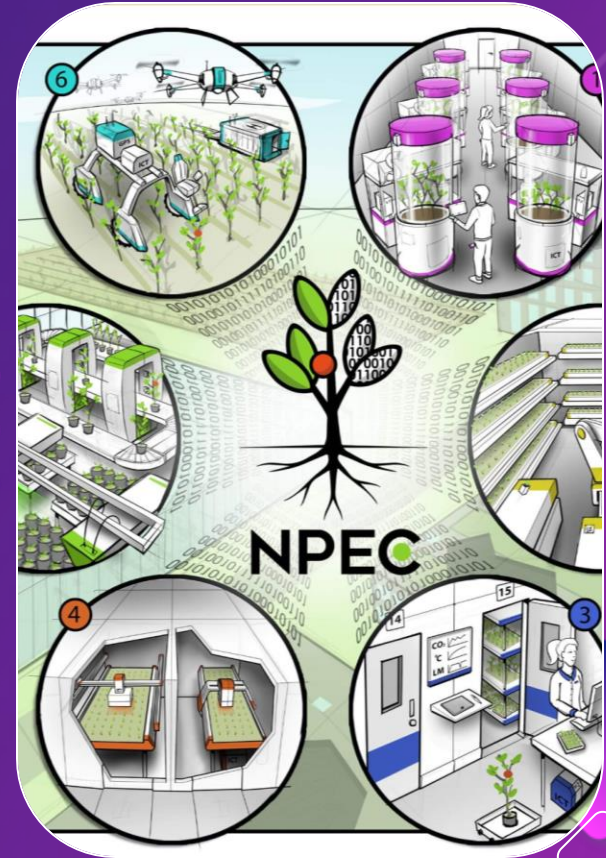
Decorative elements in the top right corner include concentric circles, a blue-to-purple gradient diamond, a white outline diamond, a blue-to-purple gradient torus, and a vertical sequence of four circles (three white, one blue).

Root Analysis

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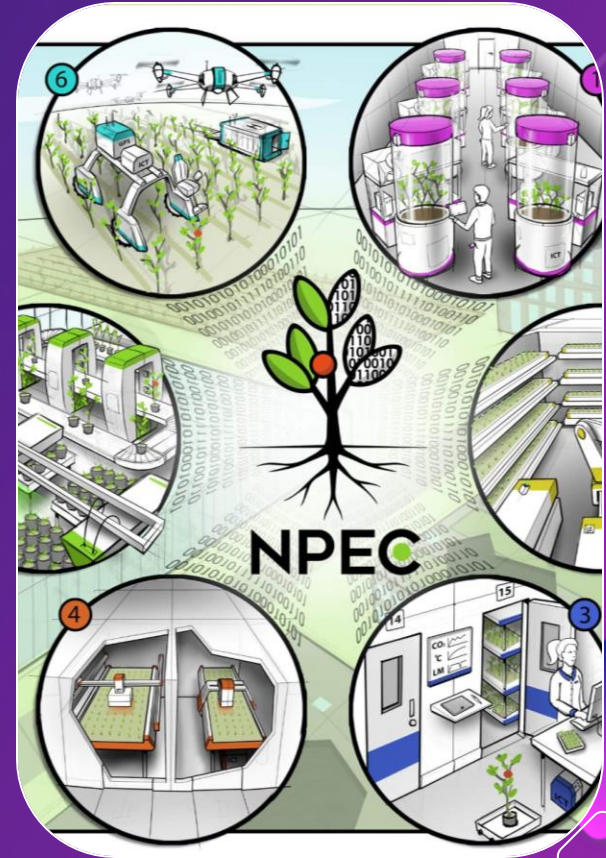
Introduction

- The Netherlands Plant Eco-phenotyping Centre (NPEC) seeks to use advanced plant phenotyping to address future food and material demands while reducing environmental impact.
- This project supports NPEC's mission by enabling precise root analysis, which enhances plant phenotyping capabilities and helps researchers identify traits for improved crop performance.



Problem definition

- **Problem:** Plant scientists face difficulties in accurately and efficiently segmenting and analyzing root systems from high-resolution images.
- **Impact:** Manual root measurements are time-consuming, prone to human error, and insufficient for high-throughput experiments.



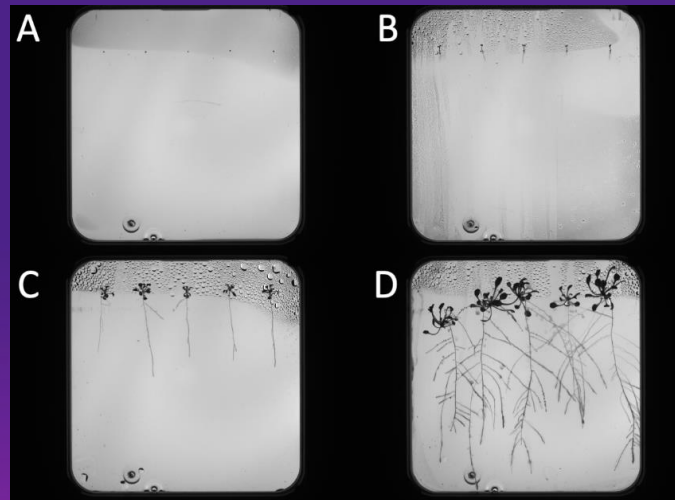
Overview of the proposed solution



My approach

Steps:

1. Preprocessing: Clean and prepare raw images.
2. Patch Creation: Divide images into smaller sections for efficient processing.
3. Model Training: Use deep learning-based segmentation to generate root masks.
4. Post-Processing: Isolate individual roots and extract features like primary root length.

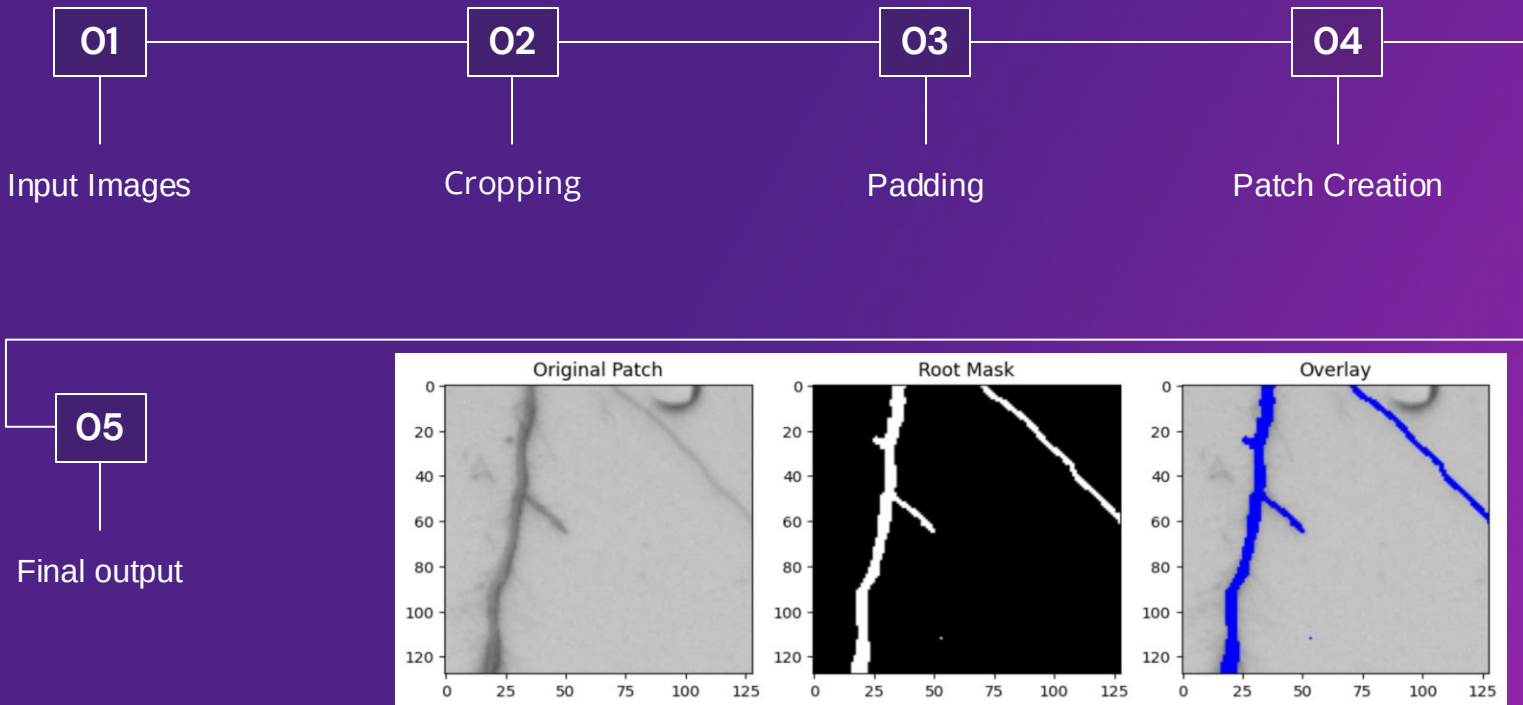


How my solution helps the client

- **Efficiency:** Automates segmentation and measurement of root systems, saving time and reducing errors.
- **Genotype Study:** Correlates plant genotypes with root growth and resilience.
- **Sustainability:** Supports research on crops with better nutrient and water uptake for sustainable agriculture.

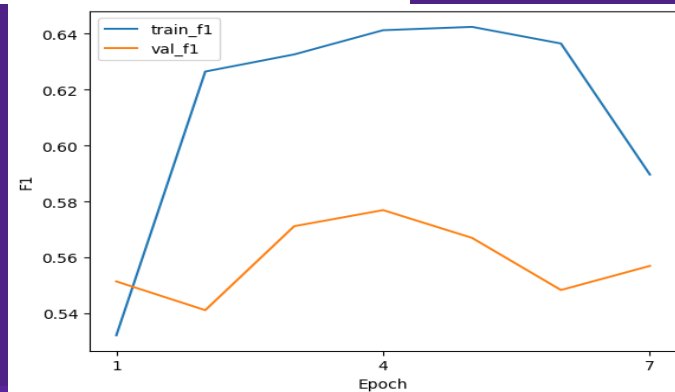
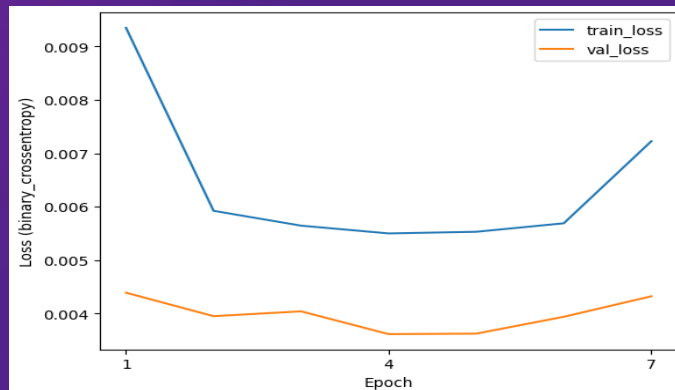
Patch Generation

This step ensures high-quality data for efficient model training



Model Training

- **Model:**
 - U-Net for generating root masks .
- **Data:**
 - 178,596 patches of images and masks.
- **Metrics:**
 - Best Validation Loss:
0.0036
 - Best Validation F1
Score: **0.577**

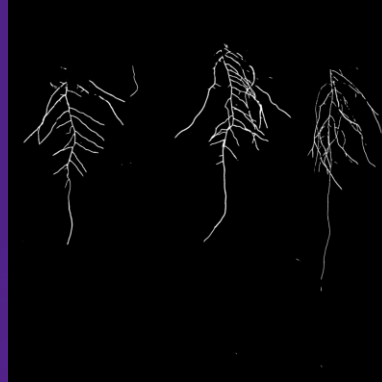


Model Inference

Pipeline:

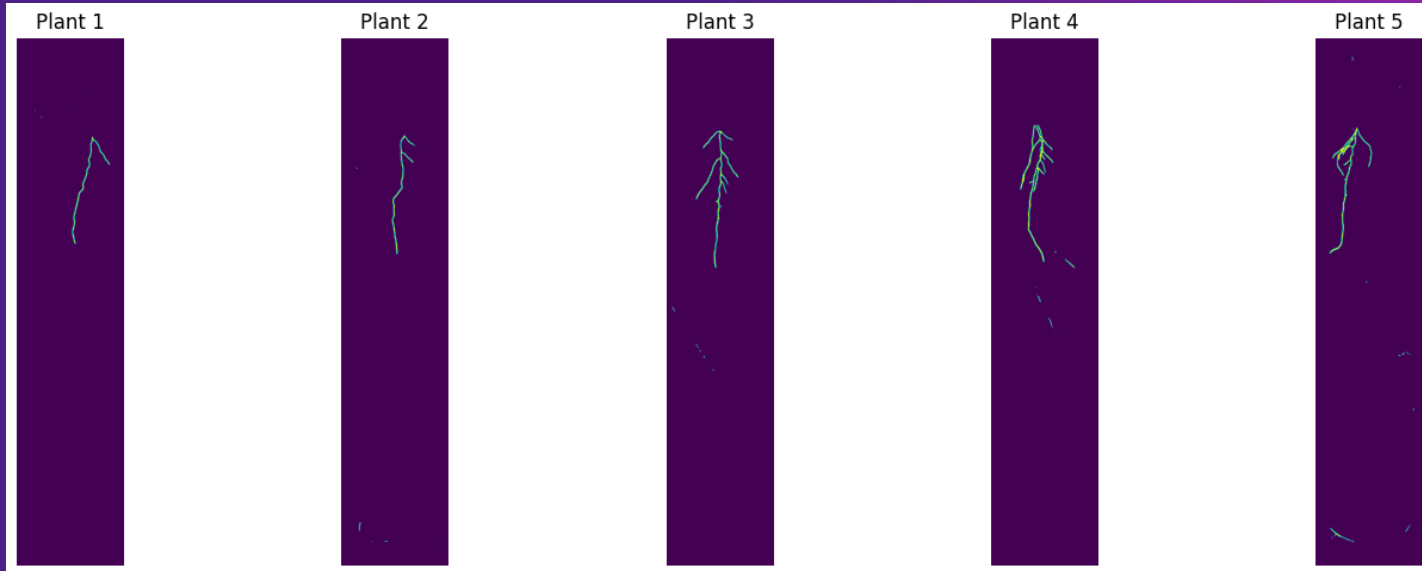


Results:



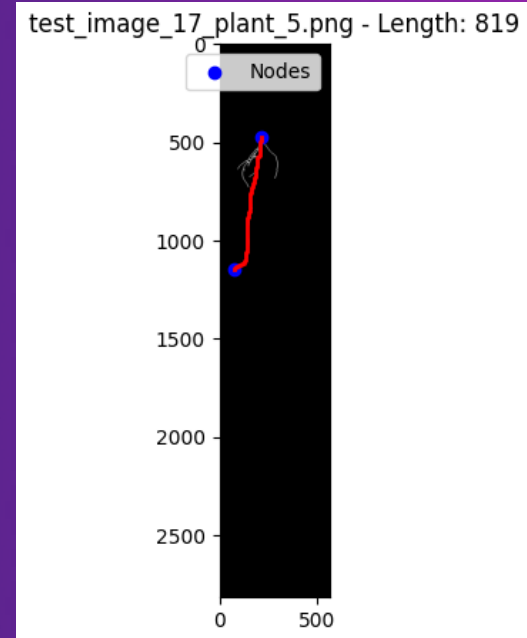
Root Segmentation and Isolation

After generating predicted masks for the Kaggle dataset, the masks were divided into five sections, based on the assumption that each mask contains five distinct roots.



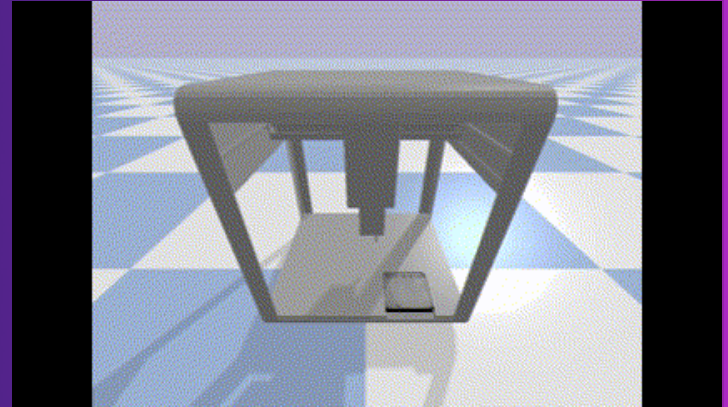
Root Length Calculation

- **Preprocessing:** Remove small regions.
- **Primary Root Isolation:** Identify the largest vertically spanning component.
- **Skeletonization:** Extract a 1-pixel-wide structure of the root.
- **Graph Construction:** Build a graph from skeletonized nodes.
- **Shortest Path Calculation:** Measure the primary root length.
- **Results:** I achieved an sMAPE score of 25.159 on the Kaggle platform, demonstrating the accuracy of my root length measurements.



Simulation of Pipette Control with PID

- **Objective:** To guide a robotic pipette to reach the target position [0,0,0.17] with high precision.
- **Accuracy:** The pipette achieved sub-millimeter (0.00098) precision using the PID controller.
- **Efficiency:** The target position was reached in just **37 steps**, demonstrating effective tuning of the controller.
- **Application:** Enables precise control for automated liquid handling in experiments.
- **PID Controller Parameters**
 - k_p : Proportional gain (40.0 for X, Y, Z axes in this code).
 - k_i : Integral gain (0 for all axes in this code).
 - k_d : Derivative gain (0 for all axes in this code).

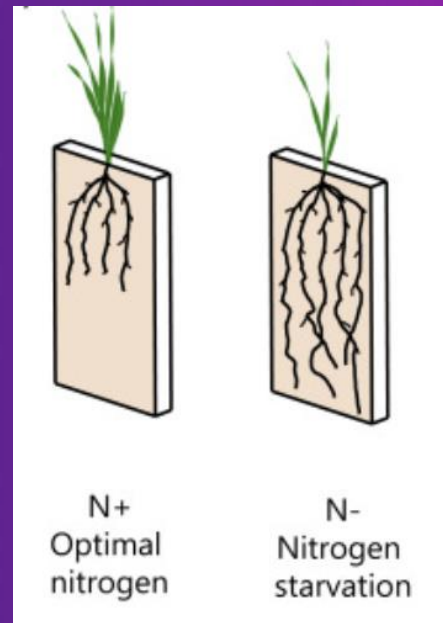


Error Analysis & Iteration

	Image Cropping and Alignment	Troubleshooting F1 Score Issues	Initial Root Length Measurement on Kaggle:
Error	Misaligned overlays and cropped regions caused inconsistencies.	Low F1 scores revealed discontinuities in predicted root masks.	Measuring root length by the longest root gave an sMAPE score of 72.
Analysis	Visualized overlays to identify misaligned masks.	Identified gaps in root skeletons.	Failed to capture the full extent of primary roots.
Solution	Refined cropping parameters and ensured proper alignment.	Used another dataset.	Measure the primary root length by the shortest path calculation.
Result	Streamlined dataset preparation, enabling accurate predictions.	Enhanced F1 scores and more accurate root length measurements.	Achieved an improved sMAPE score of 25.159.

Improvements

- **Dataset Refinement:** Improve mask segmentation and include diverse root samples.
- **Model Enhancement:** Experiment with advanced architectures and optimize hyperparameters.
- **Measurement Automation:** Automate error detection and validation for root measurements.
- **Scalability:** Adapt pipeline for large-scale plant phenotyping and build user-friendly tools.



Thank you!

